CS 210: Worksheet 14 *Dr. Mudassir Shabbir, October 13, 2026*

Student ID 1: _____ **Seat Location 1:** row:_____ col:_____

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Cartesian products, relations, and their properties. Begin with computations and finish with proofs and constructions. Show your reasoning. Prove claims or give checked counterexamples.

1. *Easy*

Let $C = \{\text{Discrete Math}, \text{Calculus}\}$ and $P = \{\text{Ali}, \text{Sara}\}$ be courses and instructors.

(a) List $C \times P$. What does an element of this product describe?

(b) Discrete Math is taught by Ali, and Calculus by both instructors. Write the teaching relation $T \subseteq C \times P$ and its inverse.

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2. *Easy*

Let A be students and B library books. Let R contain (s, b) when student s is required to read book b, and let S contain it when s has read b.

(a) Describe $R \cap S$ and $R \setminus S$ in ordinary language.

(b) Describe $S \setminus R$ and $R \triangle S$. Which relation identifies unfinished required reading?

3. *Moderate*

On $\mathbb{R}$, define $xRy \iff xy \geq 0$.

(a) Decide reflexivity, symmetry, and antisymmetry. Give proofs or numerical counterexamples.

(b) Is R transitive? Pay special attention to what happens when the middle number is zero.

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4. *Moderate*

The same empty relation can behave differently on different underlying sets.

(a) On $A = \{1, 2, 3\}$, classify $R = \varnothing$ as reflexive, symmetric, antisymmetric, and transitive. Explain using the quantified definitions.

(b) Repeat for $A = \varnothing$. Which answer changes, and why?

5. *Moderate*

A proposed argument claims that every symmetric, transitive relation R on A is reflexive: "Fix $a \in A$ and choose b with aRb. Symmetry gives bRa, so transitivity gives aRa."

(a) Identify the unsupported step. Give a nonempty relation on a nonempty set that disproves the claim.

(b) Add a hypothesis that makes this argument valid, and prove the corrected statement.

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6. *Challenge*

Let R be a relation on A, and let $\Delta_A = \{(a, a) : a \in A\}$ be the diagonal relation.

(a) Prove that R is symmetric if and only if $R = R^{-1}$.

(b) Prove that R is antisymmetric if and only if $R \cap R^{-1} \subseteq \Delta_A$.